# Revenue Management

Case 1

## 1 What factors do you think one should incorporate in the demand function? Please, discuss.

We should include several factors in a demand function. First is the current manufacture suggested retail price or MSRP of an item. The MSRP will determine the average volume of the product normally. Lower-priced items may have higher volume, while higher costs generally have lower volume. Second, we will need to know the discount percentage on the product. The discount percentage has an impact on the demand, where the higher the discount percentage the higher the demand. Third, we will need to determine the relationship between MSRP ad discount. Discounts on more expensive items may lead to a disproportionate increase in demand as compared with the same discount on lower-priced products. A simple explanation of this is that low-priced commodity items will have a stable demand regardless of price. Next, we will need to consider the time since the last promotion. There is a known stock-up effect in grocery, where the first discount period is when many consumers buy more of a product than they normally would and thus do not require that product in subsequent weeks. Next, any seasonal adjustment factors such as thanksgiving or other holidays would help to understand and factor demand, particularly seasonal adjustments. Next, we would need to know the number and performance of competitor's products that are similar. Of a similar product will be on discount in one week it will likely effect the sales of that product normally. Lastly, the demographic factors of outlying areas and their future prospects may be interesting to help predict demand stability and changes in demand due to larger macro-economic factors.

| Factors to Include | Rationale |
|---|---|
| MSRP | Need to understand price normally to determine predicted volume. High price items have low volume usually |
| Discount % | Understand what % off is being offered |
| MSRP * Discount effect | % off per price not always a linear relationship for demand. This should be accounted for. |
| Time since last promo | Consumer memory and stock up effects may affect promotion sales |
| Seasonal adjustment factors | Seasonality of products effects sales. i.e. pumpkins sold mostly in October-November |
| Competitors discount | Cannibalization of sales |
| Demographic Macro-trends | Future sales and where the trends might be leading. Paticularly relevant with COVID demographic shift. |

## 2 Estimation Single Item

Certain data cleaning and assumptions were made to facilitate the data processing. First the price column was validated to contain no 0 values. A price of $0 is possible but would inhibit the log function. For the same reason, the NA values from the previous price and trend growth were removed. As a result The first two observations were removed from the dataset. This represents less than 2% of the training data and therefore is acceptable. Additionally, since the log of the price is predicted for the accuracy and prediction values the predicted values were normalized to units not log units where possible.

a. Using the R script in the attached document and training data we are able to estimate the parameters for the following model $\log(dt) = a_0 + a_t + \text{Trend} \cdot t + \beta_0 \log(p_t)$
The model estimates result in the following parameters:

| Parameter | Name | Description | Estimate |
|---|---|---|---|
| $a_0$ | (Intercept) | Base level demand based on week_factor1f | 5.20355 |
| $a_t$ | Week_factor2 | The lift from week factor 2. 2 is approximation for month | 0.23352 |
| $a_t$ | Week_factor3 | | -0.17897 |
| $a_t$ | Week_factor4 | | -0.08299 |
| $a_t$ | Week_factor5 | | -0.09413 |
| $a_t$ | Week_factor6 | | 0.230207 |
| $a_t$ | Week_factor7 | | -0.24168 |
| $a_t$ | Week_factor8 | | -0.08354 |
| $a_t$ | Week_factor9 | | -0.18146 |
| $a_t$ | Week_factor10 | | 0.296648 |
| $a_t$ | Week_factor11 | | 0.256366 |
| $a_t$ | Week_factor12 | | -0.22154 |
| $a_t$ | Week_factor13 | | 0.029748 |
| Trend | Trend_1 | Trend is lift or % difference from previous week. | 0.002461 |
| $\beta_0$ | log(Price) | The factor that effects the log of the price. Price must be non-zero or error. | -3.69918 |

The following summary from model a was then generated.

```
Call:
lm(formula = log(St) ~ Week_factor + Week + log(Price), data =
train_data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.86995 -0.17959  0.04269  0.24223  0.78927

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.203550   0.128925  40.361   <2e-16 ***
Week_factor2   0.233520   0.174404   1.339   0.1840
Week_factor3  -0.178967   0.177538  -1.008   0.3162
Week_factor4  -0.082988   0.176317  -0.471   0.6390
Week_factor5  -0.094127   0.179483  -0.524   0.6013
Week_factor6   0.230207   0.176476   1.304   0.1954
Week_factor7  -0.241677   0.181199  -1.334   0.1857
Week_factor8  -0.083542   0.182336  -0.458   0.6479
Week_factor9  -0.181463   0.181117  -1.002   0.3191
Week_factor10  0.296648   0.181280   1.636   0.1053
Week_factor11  0.256366   0.184754   1.388   0.1687
Week_factor12 -0.221539   0.185099  -1.197   0.2345
Week_factor13  0.029748   0.186255   0.160   0.8735
Week           0.002461   0.001326   1.856   0.0667 .
log(Price)    -3.699180   0.342353 -10.805   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3486 on 89 degrees of freedom
Multiple R-squared:  0.6114,    Adjusted R-squared:  0.5503
F-statistic:    10 on 14 and 89 DF,  p-value: 5.204e-13
```

The test data was used to estimate the prediction accuracy and power of the model. The following results table was generated by predicting the test data and comparing it against the true values.

| Model Estimate Name | Value | Explanation |
|---|---|---|
| $R^2$ | 0.32 | This coefficient of determination indicates the proportion that the dependent variables can explain the independent variable. The value varies between -1 and 1. The closer to 1 the most explanation the model offers |
| MAPE | 88.87% | Mean Absolute Percent Error. This value is the average deviation or error of the predictions. In other words, on average, how wrong is the model? |
| Accuracy of Linear regression | 11.12% | This is the inverse of the MAPE and indicates on average how correct the model is |

b. Using the R script in the attached document and training data, we can estimate the parameters for the following model $\log(dt) = a_0 + a_t + \text{Trend} \cdot t + \beta_0 \log(p_t) + \beta_1 \log(p_t - 1) + \beta_2 \log(p_t - 2)$.
The model estimates result in the following parameters:

| Parameter | Name | Description | Estimate |
|---|---|---|---|
| $a_0$ | (Intercept) | Base level demand based on week_factor1f | 5.396454 |
| $a_t$ | Week_factor2 | The lift from week factor 2. 2 is approximation for month | 0.107202 |
| $a_t$ | Week_factor3 | | 0.202331 |
| $a_t$ | Week_factor4 | | 0.066698 |
| $a_t$ | Week_factor5 | | 0.204127 |
| $a_t$ | Week_factor6 | | 0.42544 |
| $a_t$ | Week_factor7 | | 0.028417 |
| $a_t$ | Week_factor8 | | 0.442834 |
| $a_t$ | Week_factor9 | | 0.221023 |
| $a_t$ | Week_factor10 | | 0.383419 |
| $a_t$ | Week_factor11 | | 0.322523 |
| $a_t$ | Week_factor12 | | 0.262324 |
| $a_t$ | Week_factor13 | | 0.015288 |
| Trend | Trend_1 | Trend is lift or % difference from previous week. | 0.000412 |
| $\beta_0$ | log(Price) | The factor that effects the log of the price. Price must be non-zero or error. | -2.92081 |
| $\beta_1$ | log(Price_1) | The factor effecting the price based on the previous week's data | 2.928279 |
| $B_2$ | log(Price_2) | The factor effecting price based on 2 weeks ago's data. | 1.894198 |

The following summary from model b was then generated.

```
Call:
glm(formula = log(St) ~ Week_factor + Week + log(Price) + log(Price_1)
+
    log(Price_2), data = train_data)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-0.24147  -0.07295  -0.00887   0.07000   0.34604

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.3964535  0.0464434 116.194  < 2e-16 ***
Week_factor2   0.1072018  0.0621134   1.726  0.08791 .
Week_factor3   0.2023309  0.0649496   3.115  0.00249 **
Week_factor4   0.0666978  0.0628632   1.061  0.29163
Week_factor5   0.2041267  0.0650093   3.140  0.00231 **
Week_factor6   0.4254402  0.0631454   6.737 1.67e-09 ***
Week_factor7   0.0284168  0.0652580   0.435  0.66431
Week_factor8   0.4428343  0.0680922   6.503 4.78e-09 ***
Week_factor9   0.2210226  0.0667118   3.313  0.00135 **
Week_factor10  0.3834187  0.0644279   5.951 5.46e-08 ***
Week_factor11  0.3225230  0.0656173   4.915 4.14e-06 ***
Week_factor12  0.2623244  0.0686746   3.820  0.00025 ***
Week_factor13  0.0152876  0.0660999   0.231  0.81764
Week           0.0004117  0.0004779   0.862  0.39127
log(Price)    -2.9208085  0.1255607 -23.262  < 2e-16 ***
log(Price_1)   2.9282787  0.1274345  22.979  < 2e-16 ***
log(Price_2)   1.8941977  0.1267821  14.941  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for gaussian family taken to be 0.01530739)

    Null deviance: 27.8400  on 103  degrees of freedom
Residual deviance:  1.3317  on  87  degrees of freedom
AIC: -122.08

Number of Fisher Scoring iterations: 2
```

The test data was used to estimate the prediction accuracy and power of the model. The following results table was generated by predicting the test data and comparing against the true values.

| Model Estimate Name | Value | Explanation |
|---|---|---|
| $R^2$ | 0.93 | This coefficient of determination indicates the proportion that the dependent variables can explain the independent variable. The value varies between -1 and 1. The closer to 1 the most explanation the model offers |
| MAPE | 11.3% | Mean Absolute Percent Error. This value is the average deviation or error of the predictions. In other words, on average, how wrong is the model? |
| Accuracy of Linear regression | 88.5% | This is the inverse of the MAPE and indicates on average how correct the model is |

## 3 For the model in equation (2), what can you say on the magnitudes of the estimated parameters $\beta_0$, $\beta_1$ and $\beta_2$? Does this make sense and why?

The magnitudes of the beta factors indicate that $\beta_0$ is the most important predictor, while $\beta_2$ and $\beta_1$ are second and third most important, respectively. In absolute terms, $\beta_0$ has a coefficient 5 times that of $\beta_1$, indicating it is 5 times as important. Similarly, $\beta_2$ is 3.6 times as important as $\beta_1$. All of the factors

have the same order of magnitude (between 100 and 800), indicating that the price factors have a proportional effect on demand. Specifically, there is no price that is an order of magnitude above the rest, indicating the significant importance on this factor compared to the others.

These factors' relative magnitude and importance make sense in a consumer or B2C setting. $\beta_2$ is a parameter estimation for the demand from 2 weeks previous. There are two primary psychological effects that explain the importance of these effects.

First, consumers are generally price sensitive and have price memory. Therefore, the price of an item at time t should have a strong relationship with the demand at time t. Furthermore, consumers have a memory of pricing that may last through time. If there is a promotion at time t then the price is lower than it was at time t, incentivizing consumers to purchase. If at time t-1 there is a promotion and at time t there is no promotion, the consumer may opt to substitute their item or wait until there is a promotion in the future to purchase.

Second, consumers often "stock up" during the first promotion or during deep dives. The stock up effect means that at time t+1 consumers have not used the product they have previously purchased. We would expect to see a dip in sales at time t+1, therefore the demand is decoupled from the price and the coefficient reduced. At time t+2 when the consumers have used their stockpile of product, they will likely need to decide whether to purchase again. The price at time t+2 is more closely related to demand than that of time t+1. This stock up effect helps explain the coefficient importance for price at t-1 and t-2.

## 4 Assume that $p_7 = 1$, $p_6 = 0.8$ and $p_5 = 0.8$. Compute the values of the predicted demand $d_7$ using models (1) and (2) above.

In order to use the new data to predict $d_7$ the model is expecting additional data that we will need to the model. Therefore, =7these will be added as assumptions. We will assume that the Week_factor is equal to 2. We will need to refactor the variable to be 2. We will also assume that the trend_1 is 0. These data are required for the model to perform optimally.

After adding these factors, model a prediction that the demand for week 7 is 164.78 units. The originally predicted value is 5.1 but this corresponds to the log of demand. Similarly, model b's output is 78.6 units or 4.36 before the exponential increase.

## 5 Discuss the differences between the two models (1) and (2) above. For which type of products do you think that model (2) is better? Please discuss.

Both models rely on different assumptions and produce varying outputs. Model 2 accounts for consumer memory by adding in the previous week's pricing, while model 1 does not. Model 2 seems to give more accurate predictions as seen via a lower MAPE. However, the choice of model may depend on the product type. *Grocery* also called fast moving consumer goods items are ones that consumers purchase regularly with little variation. The forecasts can sometimes be 90% or more accurate with medium-term stability in pricing. For these items model 2 is the most appropriate since it accounts for more real features. Model 2 accounts for consumer memory, and stockup commonly seen in the grocery-

type market to give higher accuracy. For *fashion-type* items, where the demand is highly variable and generally seasonal Model 1 may be more useful. Model 1 accounts for fewer features and better replicates the consumer's on-the-spot decision-making as it only account for the current price, and trend (as seen through inventory levels or similar purchases). Therefore model 1 may help predict the consumer demand that is highly variable and generally seasonal.

**Estimation- Multiple Items**

## 6 Price Estimation with 5 Items

| Parameter | Name | Estimate |
|---|---|---|
| $a_t$ | (Intercept) | 2.057661 |
| $a_t$ | Week_factor2 | -0.15161 |
| $a_t$ | Week_factor3 | -0.19267 |
| $a_t$ | Week_factor4 | 0.035644 |
| $a_t$ | Week_factor5 | 0.138694 |
| $a_t$ | Week_factor6 | -0.16577 |
| $a_t$ | Week_factor7 | -0.12632 |
| $a_t$ | Week_factor8 | -0.01231 |
| $a_t$ | Week_factor9 | 0.109319 |
| $a_t$ | Week_factor10 | -0.09812 |
| $a_t$ | Week_factor11 | 0.11098 |
| $a_t$ | Week_factor12 | 0.186096 |
| $a_t$ | Week_factor13 | -0.14157 |
| Trend | Week | 0.009395 |
| Indicators | Prod_2 | -0.14758 |
| Indicators | Prod_3 | -0.6301 |
| Indicators | Prod_4 | -0.51645 |
| Indicators | Prod_5 | -0.29241 |
| $\beta_0^1$ | Prod_1:log(PriceB1) | -3.3939 |
| $\beta_1^1$ | Prod_1:log(PriceB1_1) | 2.93065 |
| $\beta_2^1$ | Prod_1:log(PriceB1_2) | 1.997682 |
| $\delta_{12}$ | Prod_1:PriceB2 | 0.635165 |
| $\delta_{13}$ | Prod_1:PriceB3 | 0.32136 |
| $\delta_{14}$ | Prod_1:PriceB4 | 0.675114 |
| $\delta_{15}$ | Prod_1:PriceB5 | 0.744538 |
| $\beta_0^2$ | Prod_2:log(PriceB2) | -3.0225 |
| $\beta_1^2$ | Prod_2:log(PriceB2_1) | 2.820657 |
| $\beta_2^2$ | Prod_2:log(PriceB2_2) | 1.925881 |
| $\delta_{21}$ | Prod_2:PriceB1 | 0.554194 |
| $\delta_{23}$ | PriceB3:Prod_2 | 0.792565 |
| $\delta_{24}$ | PriceB4:Prod_2 | 0.592026 |
| $\delta_{25}$ | PriceB5:Prod_2 | 0.499789 |

| | | |
|---|---|---|
| $\beta_0^3$ | Prod_3:log(PriceB3) | -3.54936 |
| $\beta_1^3$ | Prod_3:log(PriceB3_1) | 2.683976 |
| $\beta_2^3$ | Prod_3:log(PriceB3_2) | 1.762131 |
| $\delta_{32}$ | PriceB2:Prod_3 | 0.399669 |
| $\delta_{31}$ | PriceB1:Prod_3 | 0.844784 |
| $\delta_{34}$ | PriceB4:Prod_3 | 0.71796 |
| $\delta_{35}$ | PriceB5:Prod_3 | 0.59485 |
| $\beta_0^4$ | Prod_4:log(PriceB4) | -3.86365 |
| $\beta_1^4$ | Prod_4:log(PriceB4_1) | 2.803475 |
| $\beta_2^4$ | Prod_4:log(PriceB4_2) | 2.135683 |
| $\delta_{41}$ | PriceB1:Prod_4 | 0.576479 |
| $\delta_{42}$ | PriceB2:Prod_4 | 0.810766 |
| $\delta_{43}$ | PriceB3:Prod_4 | 0.789608 |
| $\delta_{45}$ | PriceB5:Prod_4 | 0.985042 |
| $\beta_0^5$ | Prod_5:log(PriceB5) | -3.93547 |
| $\beta_1^5$ | Prod_5:log(PriceB5_1) | 2.688287 |
| $\beta_2^5$ | Prod_5:log(PriceB5_2) | 1.772619 |
| $\delta_{51}$ | PriceB1:Prod_5 | 0.994232 |
| $\delta_{52}$ | PriceB2:Prod_5 | 0.844316 |
| $\delta_{53}$ | PriceB3:Prod_5 | 0.847411 |
| $\delta_{54}$ | PriceB4:Prod_5 | 0.699741 |

```
Call:
lm(formula = log(St) ~ Week_factor + Week + Prod_1 * log(PriceB1) +
    Prod_1 * log(PriceB1_1) + Prod_1 * log(PriceB1_2) + Prod_1 *
    PriceB2 + Prod_1 * PriceB3 + Prod_1 * PriceB4 + Prod_1 *
    PriceB5 - Prod_1 - log(PriceB1) - log(PriceB1_1) - log(PriceB1_2) -
    PriceB2 - PriceB3 - PriceB4 - PriceB5 + Prod_2 * log(PriceB2) +
    Prod_2 * log(PriceB2_1) + Prod_2 * log(PriceB2_2) + Prod_2 *
    PriceB1 + Prod_2 * PriceB3 + Prod_2 * PriceB4 + Prod_2 *
    PriceB5 - log(PriceB2) - log(PriceB2_1) - log(PriceB2_2) -
    PriceB1 - PriceB3 - PriceB4 - PriceB5 + Prod_3 * log(PriceB3) +
    Prod_3 * log(PriceB3_1) + Prod_3 * log(PriceB3_2) + Prod_3 *
    PriceB2 + Prod_3 * PriceB1 + Prod_3 * PriceB4 + Prod_3 *
    PriceB5 - log(PriceB3) - log(PriceB3_1) - log(PriceB3_2) -
    PriceB1 - PriceB2 - PriceB4 - PriceB5 + Prod_4 * log(PriceB4) +
    Prod_4 * log(PriceB4_1) + Prod_4 * log(PriceB4_2) + Prod_4 *
    PriceB1 + Prod_4 * PriceB2 + Prod_4 * PriceB3 + Prod_4 *
    PriceB5 - log(PriceB4) - log(PriceB4_1) - log(PriceB4_2) -
    PriceB1 - PriceB3 - PriceB2 - PriceB5 + Prod_5 * log(PriceB5) +
    Prod_5 * log(PriceB5_1) + Prod_5 * log(PriceB5_2) + Prod_5 *
    PriceB1 + Prod_5 * PriceB2 + Prod_5 * PriceB3 + Prod_5 *
    PriceB4 - log(PriceB5) - log(PriceB5_1) - log(PriceB5_2) -
    PriceB1 - PriceB3 - PriceB2 - PriceB4, data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.42990 -0.09196 -0.00181  0.08826  0.39077

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           2.057661   0.279041   7.374 7.60e-13 ***
Week_factor2         -0.151607   0.034555  -4.387 1.42e-05 ***
Week_factor3         -0.192668   0.035795  -5.383 1.16e-07 ***
Week_factor4          0.035644   0.034183   1.043 0.297613
Week_factor5          0.138694   0.035717   3.883 0.000118 ***
Week_factor6         -0.165773   0.035170  -4.714 3.22e-06 ***
Week_factor7         -0.126322   0.037003  -3.414 0.000696 ***
Week_factor8         -0.012308   0.035436  -0.347 0.728495
Week_factor9          0.109319   0.035626   3.069 0.002276 **
Week_factor10        -0.098122   0.034818  -2.818 0.005035 **
Week_factor11         0.110980   0.036585   3.033 0.002553 **
Week_factor12         0.186096   0.037160   5.008 7.81e-07 ***
Week_factor13        -0.141571   0.038009  -3.725 0.000219 ***
Week                  0.009395   0.000270  34.802  < 2e-16 ***
Prod_2               -0.147584   0.406461  -0.363 0.716699
Prod_3               -0.630097   0.408803  -1.541 0.123916
Prod_4               -0.516447   0.392710  -1.315 0.189126
Prod_5               -0.292413   0.392427  -0.745 0.456561
Prod_1:log(PriceB1)  -3.393901   0.141843 -23.927  < 2e-16 ***
Prod_1:log(PriceB1_1)  2.930650   0.141289  20.742  < 2e-16 ***
Prod_1:log(PriceB1_2)  1.997682   0.140054  14.264  < 2e-16 ***
Prod_1:PriceB2        0.635165   0.149617   4.245 2.64e-05 ***
Prod_1:PriceB3        0.321360   0.139573   2.302 0.021748 *
Prod_1:PriceB4        0.675114   0.142524   4.737 2.88e-06 ***
Prod_1:PriceB5        0.744538   0.148558   5.012 7.67e-07 ***
Prod_2:log(PriceB2)  -3.022504   0.129384 -23.361  < 2e-16 ***
Prod_2:log(PriceB2_1)  2.820657   0.127230  22.170  < 2e-16 ***
Prod_2:log(PriceB2_2)  1.925881   0.128057  15.039  < 2e-16 ***
Prod_2:PriceB1        0.554194   0.164735   3.364 0.000831 ***
PriceB3:Prod_2        0.792565   0.139007   5.702 2.11e-08 ***
PriceB4:Prod_2        0.592026   0.143762   4.118 4.52e-05 ***
PriceB5:Prod_2        0.499789   0.152263   3.282 0.001106 **
Prod_3:log(PriceB3)  -3.549364   0.119534 -29.693  < 2e-16 ***
Prod_3:log(PriceB3_1)  2.683976   0.119080  22.539  < 2e-16 ***
Prod_3:log(PriceB3_2)  1.762131   0.125392  14.053  < 2e-16 ***
PriceB2:Prod_3        0.399669   0.158037   2.529 0.011768 *
PriceB1:Prod_3        0.844784   0.164351   5.140 4.04e-07 ***
PriceB4:Prod_3        0.717960   0.144005   4.986 8.72e-07 ***
PriceB5:Prod_3        0.594850   0.148342   4.010 7.07e-05 ***
Prod_4:log(PriceB4)  -3.863654   0.121690 -31.750  < 2e-16 ***
Prod_4:log(PriceB4_1)  2.803475   0.119358  23.488  < 2e-16 ***
Prod_4:log(PriceB4_2)  2.135683   0.120494  17.724  < 2e-16 ***
PriceB1:Prod_4        0.576479   0.165092   3.492 0.000525 ***
PriceB2:Prod_4        0.810766   0.151621   5.347 1.40e-07 ***
PriceB3:Prod_4        0.789608   0.139041   5.679 2.39e-08 ***
PriceB5:Prod_4        0.985042   0.148468   6.635 9.03e-11 ***
Prod_5:log(PriceB5)  -3.935469   0.129582 -30.371  < 2e-16 ***
Prod_5:log(PriceB5_1)  2.688287   0.125727  21.382  < 2e-16 ***
Prod_5:log(PriceB5_2)  1.772619   0.128181  13.829  < 2e-16 ***
PriceB1:Prod_5        0.994232   0.166145   5.984 4.34e-09 ***
PriceB2:Prod_5        0.844316   0.149506   5.647 2.83e-08 ***
PriceB3:Prod_5        0.847411   0.138895   6.101 2.21e-09 ***
PriceB4:Prod_5        0.699741   0.142065   4.926 1.17e-06 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1485 on 467 degrees of freedom
Multiple R-squared:   0.97,     Adjusted R-squared:  0.9666
F-statistic: 290.2 on 52 and 467 DF,  p-value: < 2.2e-16
```

| Model Estimate Name | Value | Explanation |
|---|---|---|
| $R^2$ | 0.93 | This coefficient of determination indicates the proportion that the dependent variables can explain the independent variable. The value varies between -1 and 1. The closer to 1 the most explanation the model offers |
| MAPE | 13.3% | Mean Absolute Percent Error. This value is the average deviation or error of the predictions. In other words, on average, how wrong is the model? |
| Accuracy of Linear regression | 86.7% | This is the inverse of the MAPE and indicates on average how correct the model is |

**7 Repeat the same procedure as in Question 6 using the data from the csv file "multipleitem- part2.csv". As before, submit (i) the summary of the regressions; (ii) the values of R2 and MAPE on the test set**

| Parameter | Name | Estimate |
|---|---|---|
| $a_t$ | (Intercept) | 4.735857 |
| $a_t$ | Week_factor2 | -0.0969 |
| $a_t$ | Week_factor3 | -0.14944 |
| $a_t$ | Week_factor4 | 0.049208 |
| $a_t$ | Week_factor5 | 0.155785 |
| $a_t$ | Week_factor6 | -0.12093 |
| $a_t$ | Week_factor7 | -0.09032 |
| $a_t$ | Week_factor8 | 0.015598 |
| $a_t$ | Week_factor9 | 0.109074 |
| $a_t$ | Week_factor10 | -0.06778 |
| $a_t$ | Week_factor11 | 0.143215 |
| $a_t$ | Week_factor12 | 0.225915 |
| $a_t$ | Week_factor13 | -0.09257 |
| Trend | Week | 0.009074 |
| Indicators | Prod_2 | -0.12973 |
| Indicators | Prod_3 | -0.19319 |
| Indicators | Prod_4 | -0.21599 |
| Indicators | Prod_5 | 0.161141 |
| $\beta_0^1$ | Prod_1:log(PriceB1) | -3.01609 |
| $\beta_1^1$ | Prod_1:log(PriceB1_1) | 2.772655 |
| $\beta_2^1$ | Prod_1:log(PriceB1_2) | 1.939395 |
| $\delta_{12}$ | Prod_1:PriceB2 | 0.271789 |
| $\delta_{13}$ | Prod_1:PriceB3 | 0.07133 |
| $\delta_{14}$ | Prod_1:PriceB4 | -0.07212 |
| $\delta_{15}$ | Prod_1:PriceB5 | 0.000703 |
| $\beta_0^2$ | Prod_2:log(PriceB2) | -3.23043 |
| $\beta_1^2$ | Prod_2:log(PriceB2_1) | 2.951261 |
| $\beta_2^2$ | Prod_2:log(PriceB2_2) | 2.017659 |
| $\delta_{21}$ | Prod_2:PriceB1 | -1.35333 |
| $\delta_{23}$ | PriceB3:Prod_2 | 0.049266 |
| $\delta_{24}$ | PriceB4:Prod_2 | -0.04492 |
| $\delta_{25}$ | PriceB5:Prod_2 | -0.19113 |
| $\beta_0^3$ | Prod_3:log(PriceB3) | -3.94749 |
| $\beta_1^3$ | Prod_3:log(PriceB3_1) | 2.683677 |

| | | |
|---|---|---|
| $\beta_2^3$ | Prod_3:log(PriceB3_2) | 1.853494 |
| $\delta_{32}$ | PriceB2:Prod_3 | -0.02093 |
| $\delta_{31}$ | PriceB1:Prod_3 | -1.11948 |
| $\delta_{34}$ | PriceB4:Prod_3 | 0.029043 |
| $\delta_{35}$ | PriceB5:Prod_3 | -0.26536 |
| $\beta_0^4$ | Prod_4:log(PriceB4) | -3.81274 |
| $\beta_1^4$ | Prod_4:log(PriceB4_1) | 3.006401 |
| $\beta_2^4$ | Prod_4:log(PriceB4_2) | 2.167476 |
| $\delta_{41}$ | PriceB1:Prod_4 | -0.63791 |
| $\delta_{42}$ | PriceB2:Prod_4 | -0.10041 |
| $\delta_{43}$ | PriceB3:Prod_4 | -0.04657 |
| $\delta_{45}$ | PriceB5:Prod_4 | -0.28381 |
| $\beta_0^5$ | Prod_5:log(PriceB5) | -3.95599 |
| $\beta_1^5$ | Prod_5:log(PriceB5_1) | 2.753689 |
| $\beta_2^5$ | Prod_5:log(PriceB5_2) | 2.177222 |
| $\delta_{51}$ | PriceB1:Prod_5 | -1.48997 |
| $\delta_{52}$ | PriceB2:Prod_5 | -0.10951 |
| $\delta_{53}$ | PriceB3:Prod_5 | -0.10385 |
| $\delta_{54}$ | PriceB4:Prod_5 | -0.11377 |

```
Call:
lm(formula = log(St) ~ Week_factor + Week + Prod_1 * log(PriceB1) +
    Prod_1 * log(PriceB1_1) + Prod_1 * log(PriceB1_2) + Prod_1 *
    PriceB2 + Prod_1 * PriceB3 + Prod_1 * PriceB4 + Prod_1 *
    PriceB5 - Prod_1 - log(PriceB1) - log(PriceB1_1) - log(PriceB1_2) -
    PriceB2 - PriceB3 - PriceB4 - PriceB5 + Prod_2 * log(PriceB2) +
    Prod_2 * log(PriceB2_1) + Prod_2 * log(PriceB2_2) + Prod_2 *
    PriceB1 + Prod_2 * PriceB3 + Prod_2 * PriceB4 + Prod_2 *
    PriceB5 - log(PriceB2) - log(PriceB2_1) - log(PriceB2_2) -
    PriceB1 - PriceB3 - PriceB4 - PriceB5 + Prod_3 * log(PriceB3) +
    Prod_3 * log(PriceB3_1) + Prod_3 * log(PriceB3_2) + Prod_3 *
    PriceB2 + Prod_3 * PriceB1 + Prod_3 * PriceB4 + Prod_3 *
    PriceB5 - log(PriceB3) - log(PriceB3_1) - log(PriceB3_2) -
    PriceB1 - PriceB2 - PriceB4 - PriceB5 + Prod_4 * log(PriceB4) +
    Prod_4 * log(PriceB4_1) + Prod_4 * log(PriceB4_2) + Prod_4 *
    PriceB1 + Prod_4 * PriceB2 + Prod_4 * PriceB3 + Prod_4 *
    PriceB5 - log(PriceB4) - log(PriceB4_1) - log(PriceB4_2) -
    PriceB1 - PriceB3 - PriceB2 - PriceB5 + Prod_5 * log(PriceB5) +
    Prod_5 * log(PriceB5_1) + Prod_5 * log(PriceB5_2) + Prod_5 *
    PriceB1 + Prod_5 * PriceB2 + Prod_5 * PriceB3 + Prod_5 *
    PriceB4 - log(PriceB5) - log(PriceB5_1) - log(PriceB5_2) -
    PriceB1 - PriceB3 - PriceB2 - PriceB4, data = train_data)

Residuals:
      Min        1Q    Median        3Q       Max
-0.50318 -0.08197  0.00107  0.09885  0.48520

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)            4.7358571  0.3143709  15.065  < 2e-16 ***
Week_factor2          -0.0968973  0.0389305  -2.489 0.013158 *
Week_factor3          -0.1494441  0.0403270  -3.706 0.000236 ***
Week_factor4           0.0492078  0.0385112   1.278 0.201972
Week_factor5           0.1557847  0.0402397   3.871 0.000124 ***
Week_factor6          -0.1209321  0.0396226  -3.052 0.002402 **
Week_factor7          -0.0903211  0.0416884  -2.167 0.030772 *
Week_factor8           0.0155984  0.0399226   0.391 0.696185
Week_factor9           0.1090742  0.0401369   2.718 0.006821 **
Week_factor10         -0.0677817  0.0392263  -1.728 0.084655 .
Week_factor11          0.1432154  0.0412174   3.475 0.000559 ***
Week_factor12          0.2259154  0.0418652   5.396 1.08e-07 ***
Week_factor13         -0.0925728  0.0428213  -2.162 0.031138 *
Week                   0.0090744  0.0003041  29.835  < 2e-16 ***
Prod_2                -0.1297265  0.4579238  -0.283 0.777078
Prod_3                -0.1931862  0.4605624  -0.419 0.675075
Prod_4                -0.2159878  0.4424313  -0.488 0.625649
Prod_5                 0.1611412  0.4421120   0.364 0.715664
Prod_1:log(PriceB1)   -3.0160887  0.1598019 -18.874  < 2e-16 ***
Prod_1:log(PriceB1_1)  2.7726545  0.1591779  17.419  < 2e-16 ***
Prod_1:log(PriceB1_2)  1.9393947  0.1577864  12.291  < 2e-16 ***
Prod_1:PriceB2         0.2717892  0.1685605   1.612 0.107548
Prod_1:PriceB3         0.0713299  0.1572442   0.454 0.650309
Prod_1:PriceB4        -0.0721163  0.1605695  -0.449 0.653548
Prod_1:PriceB5         0.0007026  0.1673667   0.004 0.996652
Prod_2:log(PriceB2)   -3.2304320  0.1457651 -22.162  < 2e-16 ***
Prod_2:log(PriceB2_1)  2.9512611  0.1433392  20.589  < 2e-16 ***
Prod_2:log(PriceB2_2)  2.0176586  0.1442706  13.985  < 2e-16 ***
Prod_2:PriceB1        -1.3533325  0.1855922  -7.292 1.32e-12 ***
PriceB3:Prod_2         0.0492662  0.1566069   0.315 0.753217
PriceB4:Prod_2        -0.0449164  0.1619640  -0.277 0.781654
PriceB5:Prod_2        -0.1911338  0.1715414  -1.114 0.265760
Prod_3:log(PriceB3)   -3.9474931  0.1346687 -29.313  < 2e-16 ***
Prod_3:log(PriceB3_1)  2.6836772  0.1341564  20.004  < 2e-16 ***
Prod_3:log(PriceB3_2)  1.8534940  0.1412678  13.120  < 2e-16 ***
PriceB2:Prod_3        -0.0209336  0.1780460  -0.118 0.906455
PriceB1:Prod_3        -1.1194768  0.1851590  -6.046 3.04e-09 ***
PriceB4:Prod_3         0.0290427  0.1622374   0.179 0.858005
PriceB5:Prod_3        -0.2653619  0.1671243  -1.588 0.113006
Prod_4:log(PriceB4)   -3.8127437  0.1370972 -27.811  < 2e-16 ***
Prod_4:log(PriceB4_1)  3.0064011  0.1344702  22.357  < 2e-16 ***
Prod_4:log(PriceB4_2)  2.1674764  0.1357502  15.967  < 2e-16 ***
PriceB1:Prod_4        -0.6379055  0.1859948  -3.430 0.000658 ***
PriceB2:Prod_4        -0.1004096  0.1708179  -0.588 0.556940
PriceB3:Prod_4        -0.0465725  0.1566450  -0.297 0.766360
PriceB5:Prod_4        -0.2838055  0.1672655  -1.697 0.090413 .
Prod_5:log(PriceB5)   -3.9559879  0.1459884 -27.098  < 2e-16 ***
Prod_5:log(PriceB5_1)  2.7536891  0.1416455  19.441  < 2e-16 ***
Prod_5:log(PriceB5_2)  2.1772218  0.1444103  15.077  < 2e-16 ***
PriceB1:Prod_5        -1.4899729  0.1871812  -7.960 1.32e-14 ***
PriceB2:Prod_5        -0.1095137  0.1684349  -0.650 0.515893
PriceB3:Prod_5        -0.1038487  0.1564807  -0.664 0.507241
PriceB4:Prod_5        -0.1137718  0.1600516  -0.711 0.477535
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1673 on 467 degrees of freedom
Multiple R-squared:  0.9748,    Adjusted R-squared:  0.9719
F-statistic: 346.8 on 52 and 467 DF,  p-value: < 2.2e-16
```

| Model Estimate Name | Value | Explanation |
|---|---|---|
| $R^2$ | 0.97 | This coefficient of determination indicates the proportion that the dependent variables can explain the independent variable. The value varies between -1 and 1. The closer to 1 the most explanation the model offers |
| MAPE | 14.4% | Mean Absolute Percent Error. This value is the average deviation or error of the predictions. In other words, on average, how wrong is the model? |
| Accuracy of Linear regression | 85.6% | This is the inverse of the MAPE and indicates on average how correct the model is |

**8 Assume that $p^i_6 = 1$ and $p^i_5 = 1$ for all i, and $p^2_7 = 1$, $p^3_7 = 1$, $p^4_7 = 1$, $p^5_7 = 1$. Compute the values of the predicted demands $d^1_7$, $d^2_7$, $d^3_7$, $d^4_7$, $d^5_7$ using model (3) calibrated with the data from Question 6, when $p^1_7 = 1$ and when $p^1_7 = 0.7$ (i.e., a total of 10 different values). What can you say about the effect of promoting Brand 1?**

| | $p^1_7 = 1$ | $P^1_7 = 0.7$ |
|---|---|---|
| $St_7^1$ | 77.3 | 259.4 |
| $St_7^2$ | 71.0 | 60.1 |
| $St_7^3$ | 49.3 | 38.3 |
| $St_7^4$ | 101.2 | 85.1 |
| $St_7^5$ | 158.4 | 117.5 |
| Sum | 457.2 | 560.4 |
| Weighted Sum (price * volume) | 457.2 | 482.6 |

Each product brand has a different baseline demand model, as reflected when the price for product 1 is 1. Product 1 has a demand of 77.3, making it 3rd in demand. However, when the price is reduced by 30% to 70% of the original price, the demand for product 1 increases to 259.4 or 1st in terms of demand. However, there is an average 20% decrease in demand for the other products. From a supermarket perspective, with competing products and suppliers, the overall effect on sell-through and total demand is more critical. If product 1 goes on sale by 30% of the price, other products decrease demand in unit volume by an average of 20%. Assuming the price for any item is 1, we can examine a weighted price average model to examine the overall impact on the grocery store. The weighted sum of no promotion is $457.2, while the weighted sum of a 30% discount is $482.6.

Therefore, while the volume of units sold for other products increases, the total revenue more than makes up for the volume changes.

**9 Compare the estimated parameters from Questions 6 and 7. What is the difference between the 5 brands from Question 6 and the 5 brands from Question 7? Please, discuss.**

Most of the coefficients between the two models are similar and are numerically within 1 of each other. This indicates that the two models are similar. However, there are some subtle differences based on the coefficients.

The most significant difference is in the intercept. The Intercepts of the models have a difference of 2 between the two models. This indicates that the models' baseline effect without brands or any variables is different. This will bias the dependent variables. In general, the model from question 7 predicts lower importance on the indicator variable of price, which is a way to benchmark each product against the other. In other words using brand 1 as a baseline, the indicator variables adjust the prediction relative to brand 1. Additionally, there appears to be more interaction between all products and products 3 and 4 in the second

model based on the interaction term, which is much lower for these terms than for any other coefficient in the model. Therefore, the model from question 7 emphasizes the interaction of these products over the model from question 6. All other coefficients have slight differences between coefficients, which are related to different datasets and the variations with this dataset. The noise or random variations will smooth out with a larger dataset, and the coefficients will remain constant.

| Parameter | The difference in coefficient estimation fomr questions 6 to 7. Each value is absolute before subtraction. |
|---|---|
| $a_t$ | 2.678196 |
| $a_t$ | -0.05471 |
| $a_t$ | -0.04323 |
| $a_t$ | 0.013564 |
| $a_t$ | 0.017091 |
| $a_t$ | -0.04484 |
| $a_t$ | -0.036 |
| $a_t$ | 0.003288 |
| $a_t$ | -0.00024 |
| $a_t$ | -0.03034 |
| $a_t$ | 0.032235 |
| $a_t$ | 0.039819 |
| $a_t$ | -0.049 |
| Trend | -0.00032 |
| Indicators | -0.01785 |
| Indicators | -0.43691 |
| Indicators | -0.30046 |
| Indicators | -0.13127 |
| $\beta_0{}^1$ | -0.37781 |
| $\beta_1{}^1$ | -0.158 |
| $\beta_2{}^1$ | -0.05829 |
| $\delta_{12}$ | -0.36338 |
| $\delta_{13}$ | -0.25003 |
| $\delta_{14}$ | -0.60299 |
| $\delta_{15}$ | -0.74384 |
| $\beta_0{}^2$ | 0.20793 |
| $\beta_1{}^2$ | 0.130604 |
| $\beta_2{}^2$ | 0.091778 |

| | |
|---|---|
| $\delta_{21}$ | 0.799136 |
| $\delta_{23}$ | -0.7433 |
| $\delta_{24}$ | -0.54711 |
| $\delta_{25}$ | -0.30866 |
| $\beta_0^3$ | 0.39813 |
| $\beta_1^3$ | -0.0003 |
| $\beta_2^3$ | 0.091363 |
| $\delta_{32}$ | -0.37874 |
| $\delta_{31}$ | 0.274696 |
| $\delta_{34}$ | -0.68892 |
| $\delta_{35}$ | -0.32949 |
| $\beta_0^4$ | -0.05091 |
| $\beta_1^4$ | 0.202926 |
| $\beta_2^4$ | 0.031793 |
| $\delta_{41}$ | 0.061431 |
| $\delta_{42}$ | -0.71036 |
| $\delta_{43}$ | -0.74304 |
| $\delta_{45}$ | -0.70123 |
| $\beta_0^5$ | 0.02052 |
| $\beta_1^5$ | 0.065402 |
| $\beta_2^5$ | 0.404603 |
| $\delta_{51}$ | 0.495738 |
| $\delta_{52}$ | -0.73481 |
| $\delta_{53}$ | -0.74356 |
| $\delta_{54}$ | -0.58597 |

## Optimization Formulation

**10 Once the demand models are estimated from data, the next step is to formulate the optimization problem, i.e., maximizing an objective function subject to some constraints. Consider deciding promotions for 5 items throughout the next quarter (composed of 13 weeks). What types of objectives would you consider? Please discuss.**

There are several objective functions we might want to consider:

- Maximizing the sell-through of all products.
- Maximizing the profit as the difference between costs and price times volume
- Minimizing the number of promotions
- Maximizing the sell-through for one product
- Maximizing the profit of one product.

Each objective function will have benefits and drawbacks based on the business context. For instance, the business may not want to optimize the number of promotions as that is akin to having as many promotions as possible and will likely decrease revenue from the store.

Maximizing the sell-through of the products would reduce waste and likely lead to high revenue. In a grocery store setting, this objective function is useful and would result in the least waste. Furthermore, maximizing the sell-through would have to account for product interaction based on price, as some products' demand may decrease due to other product interactions.

The sell-through, however does not account for the price of the products, instead maximizing revenue would account for the price of an item and help to grow the revenue of the business. The model would optimize the number of promotions and types based on interaction terms.

However, one of the most critical considerations is maximizing the profit since the revenue is only sometimes directly proportional to the profitability of an item. This would likely have the model discounting or relying on one brand above the rest. This would likely create preference in the product offerings and unfairness in the grocery selection. Therefore this model would need to be constrained to provide equal discounts for each brand to ensure marketplace fairness.

**11 Very often, when formulating an optimization problem, it is convenient to use binary decision variables (variables that can take only the values 0 or 1). Denote the price of a particular SKU during week t by $p_t$. In addition, the (normalized) price of the SKU can only take of the following four values: 1 (i.e., full price), 0.9 (10% promotion), 0.8 (20% promotion) and 0.7 (30% promotion). How can one represent such a restriction using linear constraints and binary variables? (Hint: You may use more than one constraint.)**

We can write these constraints in the following way:

First, we define an objective function that will maximize the revenue generated

*Obj function*

$$max \sum_{t=1}^{T} \sum_{1}^{j} b_t^j * \left(p_t * \left(1 - disc_t^j\right)\right) * d_t$$

*s.t.*

$$\sum_{1}^{j} b_j \leq 1 \ \forall t$$

*Only one price per period. The binary variable that affects the discount is only allowed to be 1 all other price points for the period t will go to 0.*

$$disc_j = \in \{0, 0.1, 0.2, 0.3\}$$

$$b_j \in \{0,1\}$$

Where $dist_j^t$ is the discount on the price within the list of 0, 10%, 20% or 30% for the period t. $d_t$ is the demand at the time t. $b_j$ is a boolean or binary decision variable that controls discount j. $p_t$ is the price at time t.

**12 As we discussed in the case study, promotions should satisfy several business rules. One such rule is to limit the frequency of promotions for a particular SKU. For example, a retailer may insist that a particular product is promoted at most 4 times during the quarter. How can you model this business rule as a linear constraint with binary variables?**

For this business rule we will need to consider the time factor t for each variable.

We can write these constraints in the following way:

First, we define an objective function that will maximize the revenue generated

*Obj function*

$$max \sum_{t=1}^{T} \sum_{1}^{j} b_t^j * \left(p_t * \left(1 - disc_t^j\right)\right) * d_t$$

*s.t.*

$$\sum_{1}^{j} b_t^j \leq 1 \ \forall \ t$$

*Only one promotion per period*

$$\sum_{t=1}^{T} \sum_{1}^{j} b_t^j \leq 4$$

*4 proportions for all time periods in question*

$$disc_t^j = \in \{0, 0.1, 0.2, 0.3\}$$

$$b_t^j \in \{0,1\}$$

Where $dist_j^t$ is the discount on the price within the list of 0, 10%, 20% or 30% for the period t. $d_t$ is the demand at the time t. $b_j$ is a boolean or binary decision variable that controls discount j. $p_t$ is the price at time t.

## 13 Consider the problem of maximizing the total profit throughout the next quarter for a particular SKU (e.g., SKU88). Please, formulate the optimization problem with the constraint on the limitation of the number of promotions to be at most 4.

First, we define an objective function that will maximize the revenue generated

*Obj function*

$$max \sum_{t=1}^{T} \sum_{1}^{j} b_t^j * \left(p_t * \left(1 - disc_t^j\right) - c_t\right) * d_t$$

Profit= promotion_boolean * (price* (1-discount) -cost) *demand

*s.t.*

$$\sum_{1}^{j} 1 - d_t^j \leq 1 \ \forall \ t$$

*Only one promotion per period*

$$\sum_{t=1}^{T} \sum_{1}^{j} b_t^j \leq 4$$

*4 proportions for all time periods in question*

$$\log(demand) = \alpha_0 + a_t + Trend * t + \beta_0 * \log(p_t) + \beta_1 * \log(p_{t-1}) + \beta_2 * \log(p_{t-2})$$

*Example of demand definition. Equation 2 from case study document*

$$d_t^j = \in \{0, 0.1, 0.2, 0.3\}$$

$$b_t^j \in \{0,1\}$$

Where $dist_j^t$ is the discount on the price within the list of 0, 10%, 20% or 30% for the period t. $d_t$ is the demand at the time t. $b_j$ is a boolean or binary decision variable that controls discount j. $p_t$ is the price at time t. $c_t$ is the cost of an item at time t.

## 14 Assume that we solved the optimization problem and obtained the suggested prices for SKU88 for the next 13 weeks. The suggested prices are presented in Exhibit 5 of the case study. What can you say on this output? What tests would you conduct in order to check that the suggested prices are robust? Please discuss.

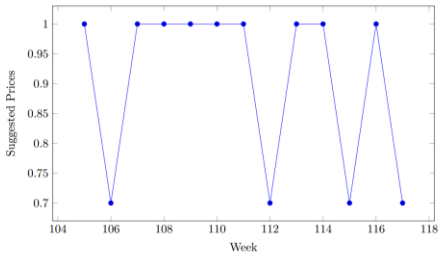



**Exhibit 5:** Suggested prices for one SKU for the coming quarter

This optimized model has several interesting output features. First the model only uses the 1 and 0.7 price points. While other price points like 10% and 20% off are available the 30% off appears to maximize the profit of the grocery store. This may be because the additional demand to price discount function follows an exponential trend. Additionally, the demand estimation for this model takes into account the memory clients' memory is generally separation between successive promotions. Promotions are at weeks 106, 112, 115, and 117 indicating at least 1 week between promotions which is typical of an objective function with memory as the first promotion is generally a "stockup" event with promotions in successive weeks returning less. Therefore, the model implements a "cooldown" period for promotions.

To ensure the model results are robust I would add several constraints and investigate the output. First, I would add a constraint to ensure that the promotions must avoid successive weeks. This should help avoid the pitfall of successive promotions with the "stock up" mentality. Furthermore, I would add a constraint to force the model to change and have at least one promotion for each price level. These constraints and model outputs would give insights into other potential optimal solutions or other scenarios which may not be optimal but would be useful to compare the model outputs to. Lastly, I would re-run the model several times and average the results and decisions. The law of large numbers indicated that with larger dataset, the trends underlying truth are better represented. By running the model several times with randomized demand data we can ensure a more robust solution.

## 15 Consider the optimization problem for several items simultaneously. Should the promotions of the different items be at the same time, or at different times? Please discuss the various factors that may affect the answer to this question. (This question is qualitative and you are not expected to solve any optimization problem.)

The optimization should limit the number of simultaneous promotions where possible. There are several reasons to limit simultaneous promotions.

First, the retailer should aim for fairness within the competitive grocery store context. Fairness in this context is avoiding the appearance of favoritism by selecting only one brand for promotion while putting several others on promotion simultaneously to compete against each other.

Second, there may be cannibalization effects with multiple promotions. Based off of the coefficients for models 6 and 7 the interaction terms indicate that as the price of other products decreases the demand for other products decreases. Therefore, when multiple items are on sale the demand for other products decreases. Intuitively the competition increases and can reduce the overall profitability of the grocery store.

Third, if the grocery store regularly has many items on promotion together, consumers may get used to large sales of products. This may damage the grocery store's reputation or have unintended consequences on the consumers such as confusing them with abundance of discounts or having them rely on massive discounts. One consequence of multi-product discounts is stockups there may be consumers who purchase many items increasing sales temporarily but decreasing demand for several weeks, and take several weeks to recover fully.

Lastly, simultaneous promotions do not allow for easy product differentiation from a consumer perspective. Products are generally differentiated by features, and price. If the price is not a differentiator because the products are all simultaneously on promotion, then consumer may not try different products or change habits. Pricing and discounts are useful for consumer migration or change between products. If the price or discounts are all aligned then there is not sufficient differentiation of products and consumer will likely not change habits therefore removing the benefit of any 'edge' customer who can change habits based on price or preference.